HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented low-level drivers for **SPI**, **I2C**, and **UART** peripherals on a **low-power microcontroller**, optimizing interrupt handling for **real-time data acquisition**.
- Developed and maintained **bootloaders** and implemented **over-the-air (OTA) firmware update mechanisms** for secure and reliable device software deployment.
- Contributed to **board bring-up** activities, integrating new hardware revisions with firmware, and debugging issues using **JTAG/SWD** interfaces and **oscilloscopes**.
- Authored comprehensive **firmware documentation** in line with **IEC 62304** standards, detailing requirements, architecture, design specifications, and verification records.
- Wrote **unit tests** and developed **Python-based automated test scripts** to verify firmware functionality and system-level integration between the device and mobile application.
- Collaborated with cross-functional teams, including systems, quality, and clinical engineers, to ensure **embedded software** met **ISO 13485** and **ISO 14971** compliance requirements.
- Integrated **Bluetooth Low Energy (BLE)** communication protocols for device-to-mobile app connectivity, ensuring efficient and secure data exchange.

SKILLS

Languages – C, C++, Python, SQL

Embedded Systems – Microcontrollers, RTOS, Bootloaders, Drivers, BSP, ARM Cortex, Low-Power Design, JTAG/SWD

Communication Protocols – BLE, SPI, I2C, UART, USB

Development Tools & Methodologies – CI/CD, Jenkins, Git, CMake, GCC, Agile/Scrum, Jira, Make

Standards & Compliance – IEC 62304, ISO 13485, ISO 14971, Embedded Security

Debugging & Testing – Oscilloscopes, Logic Analyzers, Unit Testing, Test Automation, Python Scripting

PROJECTS

Embedded BLE Controller Firmware Development

- Designed and implemented the core firmware for a **BLE controller**, focusing on ultra-low power consumption for a **battery-powered device**.
- Developed robust **low-level drivers** for peripherals such as **SPI** and **ADC**, ensuring reliable data capture for sensor input.
- Integrated secure **over-the-air (OTA) firmware update** capabilities, allowing for seamless and safe software revisions in the field.

Medical Device Software Compliance Suite

- Developed a suite of **Python scripts** to automate the generation of traceability matrices linking requirements to test cases, supporting **IEC 62304** compliance.
- Created automated **unit and integration tests** for embedded C modules using a Python framework, achieving **85% code coverage** for critical firmware components.
- Assisted in the implementation of risk management activities by documenting potential hazards and mitigation strategies as per **ISO 14971**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)